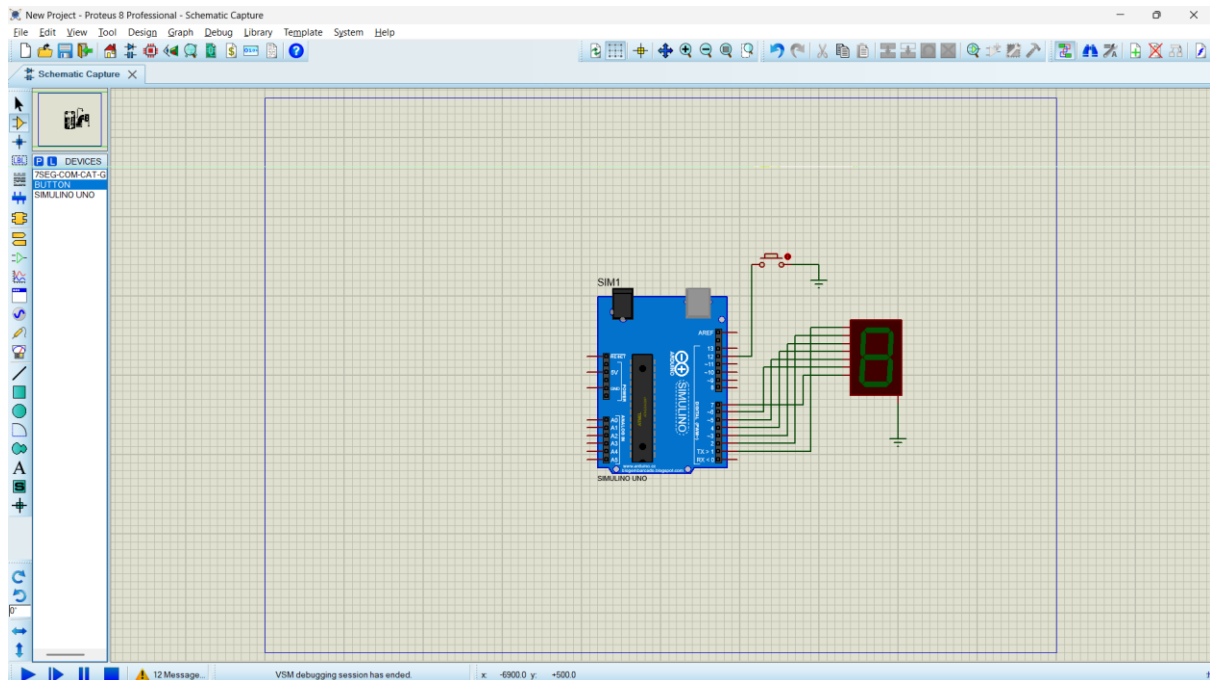


## EXP: 5 - ARDUINO BASED MANUAL ELECTRONIC COUNTER

### 1) Before Simulation



### 2) Source Code

```
int x0,x1,x2,x3,x4,x5,x6,x7,x8,x9;
```

```
int delay_time=200;
```

```
void setup() {
```

```
//configure pin2 as an input and enable the internal pull-up resistor
```

```
pinMode(12, INPUT_PULLUP);
```

```
pinMode(1,OUTPUT);
```

```
pinMode(2,OUTPUT);
```

```
pinMode(3,OUTPUT);
```

```
pinMode(4,OUTPUT);
```

```
pinMode(5,OUTPUT);
```

```
pinMode(6,OUTPUT);
```

```
pinMode(7,OUTPUT);
```

```
}
```

```
void loop() {  
  //read the pushbutton value into a variable  
  int sensorVal = digitalRead(12);  
  
  if (sensorVal == LOW) {  
    x0=true;  
  }  
  while(x0){  
    zero();  
    sensorVal = digitalRead(12);  
    if (sensorVal == LOW) {  
      x1=true;  
      x0=false;  
    }  
  }  
  while(x1){  
    one();  
    sensorVal = digitalRead(12);  
    if (sensorVal == LOW) {  
      x2=true;  
      x1=false;  
    }  
  }  
  while(x2){  
    two();  
    sensorVal = digitalRead(12);  
    if (sensorVal == LOW) {  
      x3=true;  
      x2=false;  
    }  
  }  
  while(x3){  
    three();
```

```
sensorVal = digitalRead(12);
if (sensorVal == LOW) {
  x4=true;
  x3=false;
}
}
while(x4){
  four();
  sensorVal = digitalRead(12);
  if (sensorVal == LOW) {
    x5=true;
    x4=false;
  }
}
while(x5){
  five();
  sensorVal = digitalRead(12);
  if (sensorVal == LOW) {
    x6=true;

    x5=false;
  }
}
while(x6){
  six();
  sensorVal = digitalRead(12);
  if (sensorVal == LOW) {
    x7=true;
    x6=false;
  }
}
while(x7){
  seven();
  sensorVal = digitalRead(12);
```

```

if (sensorVal == LOW) {
x8=true;
x7=false;
}
}
while(x8){
eight();
sensorVal = digitalRead(12);
if (sensorVal == LOW) {
x9=true;
x8=false;
}
}
while(x9){
nine();
sensorVal = digitalRead(12);
if (sensorVal == LOW) {
x0=true;
x9=false;
}
}

}

void zero()
{
for(int i=1;i<7;i++)
{
digitalWrite(i,HIGH);
digitalWrite(7,LOW);
}
delay(delay_time);
}

```

```
void one()
{
digitalWrite(1,LOW);
digitalWrite(2,HIGH);
digitalWrite(3,HIGH);
digitalWrite(4,LOW);
digitalWrite(5,LOW);
digitalWrite(6,LOW);
digitalWrite(7,LOW);
delay(delay_time);
}
```

```
void two()
{
digitalWrite(1,HIGH);
digitalWrite(2,HIGH);
digitalWrite(3,LOW);
digitalWrite(4,HIGH);
digitalWrite(5,HIGH);
digitalWrite(6,LOW);
digitalWrite(7,HIGH);
delay(delay_time);
}
```

```
void three()
{
digitalWrite(1,HIGH);
digitalWrite(2,HIGH);
digitalWrite(3,HIGH);
digitalWrite(4,HIGH);
digitalWrite(5,LOW);
digitalWrite(6,LOW);
digitalWrite(7,HIGH);
delay(delay_time);
}
```

```
}
```

```
void four()
```

```
{
```

```
digitalWrite(1,LOW);
```

```
digitalWrite(2,HIGH);
```

```
digitalWrite(3,HIGH);
```

```
digitalWrite(4,LOW);
```

```
digitalWrite(5,LOW);
```

```
digitalWrite(6,HIGH);
```

```
digitalWrite(7,HIGH);
```

```
delay(delay_time);
```

```
}
```

```
void five()
```

```
{
```

```
digitalWrite(1,HIGH);
```

```
digitalWrite(2,LOW);
```

```
digitalWrite(3,HIGH);
```

```
digitalWrite(4,HIGH);
```

```
digitalWrite(5,LOW);
```

```
digitalWrite(6,HIGH);
```

```
digitalWrite(7,HIGH);
```

```
delay(delay_time);
```

```
}
```

```
void six()
```

```
{
```

```
digitalWrite(1,HIGH);
```

```
digitalWrite(2,LOW);
```

```
digitalWrite(3,HIGH);
```

```
digitalWrite(4,HIGH);
```

```
digitalWrite(5,HIGH);
```

```
digitalWrite(6,HIGH);
```

```
digitalWrite(7,HIGH);
```

```
delay(delay_time);  
}
```

```
void seven()  
{  
digitalWrite(1,HIGH);  
digitalWrite(2,HIGH);  
digitalWrite(3,HIGH);  
digitalWrite(4,LOW);  
digitalWrite(5,LOW);  
digitalWrite(6,LOW);  
digitalWrite(7,LOW);  
delay(delay_time);  
}
```

```
void eight()  
{  
digitalWrite(1,HIGH);  
digitalWrite(2,HIGH);  
digitalWrite(3,HIGH);  
digitalWrite(4,HIGH);  
digitalWrite(5,HIGH);  
digitalWrite(6,HIGH);  
digitalWrite(7,HIGH);  
delay(delay_time);  
}
```

```
void nine()  
{  
digitalWrite(1,HIGH);  
digitalWrite(2,HIGH);  
digitalWrite(3,HIGH);  
digitalWrite(4,HIGH);
```

```

digitalWrite(5,LOW);

digitalWrite(6,HIGH);

digitalWrite(7,HIGH);

delay(delay_time);

}

```

### 3) Simulation After Running the code

